EGB220 Sem1_2026: PCB Design, Manufacture and Submission Guidelines

These notes are based on essential PCB Design Guidelines, prepared by Steven Bulmer and the Tech Services team. They include PCB design tools, handy hint for prototyping PCBs, and submission guidelines for the scheduled PCB runs for the unit (once a week from week 3-10). Additional guidelines are given for those team who may wish to order their own (not essential).

Choosing a PCB Design Tool

Rather than push a tool, we would like this document to suggest how best to go about making a PCB, using the QUT submission system and then sent to an external supplier.

Altium - https://www.altium.com

> Altium is the industry leading package, giving you the best foundation for a career in Electronics and or PCB design.
>
> At the time of writing, QUT has 30 seats available for using the unrestricted tool, but students must be on campus or VPN into the network in order to see the license server.
>
> Free student licenses are available for the duration of your studies at https://www.altium.com/solutions/academic-programs/student-licenses . I've not used this myself so unsure if any restrictions are in place but Im sure it would be a viable solution for most students projects.

KiCad - https://kicad.org

> A very capable free and open-source suite of tools, which most companies are adopting over the paid solutions. Digikey is a very large distributor of components who have a freely downloadable library of components to use in KiCad https://www.digikey.com.au/en/resources/design-tools/kicad

Eagle - https://www.autodesk.com/products/eagle/overview
Fusion 360 - https://www.autodesk.com/au/products/fusion-360/overview

> Note that Eagle is being phased out and replaced with Fusion 360. Bought by Autodesk and added to its suite of tools. Very strong links to its 3D design software for added 3D models of components onto your PCB design. Very useful for building boards into cases etc. It is free for students (using educational email address).

All the tools above have official training videos on their own sites in order to get up to speed with using the tool. Searching on YouTube for specific tools and tasks will yield thousands more.

PCB Design

This document sticks to a basic 2-layer board as this would be more than adequate for the majority of student designs. More layers can easily be explored but the hope is that if you go this route, you are already familiar with PCB design.

For units like EGB220 and EGB320, you will need to stick to a 2-layer board, as these are grouped together at various deadlines and submitted to a manufacturer as one batch. Students not following the rules at these deadlines will almost surely have their board rejected, so as not to delay the other groups.

Most tools have a way of creating a new PCB project that is a container for the files we create, keeping the schematic and PCB layouts together and linked so changes in one can be reflected in the other.

Schematic Capture

All PCB designs start with inputting the schematic of your circuit into the tool. All tools have a preinstalled library of components which is very handy and makes placing resistors, capacitors, power rails very easy and quick to do.

Other libraries can be easily downloaded and added to increase options within the tool, to add microcontrollers, motor drivers etc.

As a last resort, customised schematic and footprint symbols can be created in order to add any device not included in the above libraries.

Once a schematic is captured, it's a good idea to sit back, relax and have others review your work, especially when working in a team. Receiving a non-functional board from a manufacture should not be blamed on the PCB designer but on the team reviewing the work.

Good tips for new designs

- Add fuses
  - If you know a few components can only take a specific current then stick an appropriate fuse in the power path to it. Suggest a 1206 footprint to cover most fuse sizes. Can ultimately short it out if not needed, or take it off to connect in an ammeter to review current usage, or remove completely to isolate that part of the circuit from others if things not working out.
- Add LEDs on power supplies
  - Very quick immediate indication that power has come up correctly on your board
- Add test points
  - Can be as simple as a via or a pad, but somewhere you can easily get a scope into your design to check voltages.
- Add test jumpers
  - As per fuses can be used to cut a track to measure current externally
  - Can be used to switch areas between one or two voltages
  - Can be used to enable or disable certain parts of your circuit
- Add zero Ohm (0R) resistors between stages
  - Makes it easy to separate blocks for testing in isolation
- Review schematics
- Review footprint

PCB Routing

A PCB consists of many concepts which we will try to address here.

Layers

Mechanical (GM1 - Gerber Mechanical 1)

Gives the physical dimensions of the board and shows the outline that will be cut out by the manufacturer.

Copper (GTL and GBL – Gerber Top Layer and Gerber Bottom Layer)

Your board will consist of two layers, top and bottom which will have the copper tracks laid onto them in order to build up your circuit.

Silkscreen (GTO and GBO – Gerber Top Overlay and Gerber Bottom Overlay)

Overlay or silkscreen layers provide, among other things, the text designators for your components on the top and bottom layers.

Ensure there is some way to identify your board on the top overlay of your design. Good practice to have a version number in the filename and have this identified on the board too.

Solder Mask (GTS and GBS – Gerber Top Silkscreen and Gerber Bottom Silkscreen)

Normally green, this is a coating on each side of the board that prevents solder from flowing between pins and debris from shorting out tracks on a board.

Solder Paste (GTP and GBP – Gerber Top Paste and Gerber Bottom Paste)

Solder paste is a material used to assist in the soldering of surface mount components on a PCB

For the simpler designs, use the mask and paste layers that are output from the tool as is. Concentrate on the copper, silkscreen and outline of your design.

Vias

Holes in the board, plated with metal to produce an electrical connection between the top and bottom layers

Pads

Areas to place components onto a board and to apply solder to electrically connect them.

Tracks

Copper strips that connect parts of your circuit together. The width of the track determines how much current the track can carry. PCB track width calculators exist to help work out minimum widths, some exist within the tools themselves, but as a guide a 20mil wide track can carry 1A on a standard board with a 20°C temperature rise. I would suggest that for beginners in PCB design, try to keep the tracks at 20mil or above.

Gerbers

Gerber files are a recognised format output from all PCB tools that allow your design to be sent to a manufacturer for production. The tool will generate a handful of these files and they will generally need to be zipped up into a folder and sent as one to either QUT for batch production or to an external manufacturer. Some PCB design tools automatically zip the required files together on export.

Some tools generate a drill file separately to the gerber production. If so, generate the drill file and include in the zip file along with the gerbers.

Checking of Gerbers

- Use free Gerber viewers online to visualise the PCB as will be manufactured.
- Create an account with PCBWay, Submit the board and have it checked. Allows the student to go through the online checking process ensuring a good design before submitting to the QUT systems.
- Can you see all layers, holes as you had planned ?

Note : Online Gerber checking or DRC checking within the PCB tool does not check functionality. It just checks the design can be produced. It is still up to the designer and review team to check the design meets specifications and is wired accordingly.

QUT Submission

- Zip each PCB into separate folders. Each folder needs to contain all the Gerber and drill files
- Have only one PCB per Zip file
- Include your unit, team number, revision version and quantity in the filename
  - e.g. EGB220_team01_rev01_x2.zip
  - Also add your Team number to top overlay on the PCB as well.
- Email all the zip files to foe.eei@qut.edu.au before the deadline (see Canvas for dates)

PCB Arrives

- Visual inspection of the board
- Check continuity between power connections and nothing shorting to ground
- Check components against footprints
- Do you have all the required components?
- Are there specific components or areas of the board you want to test in isolation before populating the full board?

Powering Up

- Visually inspect the board
  - All components present and correct
  - All components orientated correctly
  - Check for dry or bad solder joints
  - Check for bridges between components
- Guestimate what current should be drawn from supply
- Use a good PSU with current limit for initial testing
- Ensure all power LEDs come on
- Any issues with current draw?
- Any signs of heat?

Golden Rules

- Complete the schematic first and have the team review it
- Keep tracks at 20mil (~0.5mm) width or above
- Add team number to the top overlay for easy identification once the boards arrive at QUT
- Generate all Gerbers and do not forget a drill file
- Check your Gerbers with online checkers or PCBWay before submitting your design.

PCBWay Instructions

PCBWay.com are the EEI Technicians preferred supplier of high quality, quick turnaround, and low quantity boards. Others are cheaper but some have to wait for a full panel to be populated before manufacturing starts. This could be a matter of hours or up to weeks.

Students are encouraged to make an account with PCBWay, as this can be used to check that the board they have designed is manufacturable before submitting to foe.eei@qut.edu.au for a unit batch production.

Submitting a design to PCBWay will initiate a PCB Production test to be run on it, which can take a matter of hours. After which you should get a report back about your design. At this point students can submit to foe.eei@qut.edu.au to have their board made by QUT or if no runs are left available for the unit, the students can opt to pay for the boards themselves.

There are a few things to note though

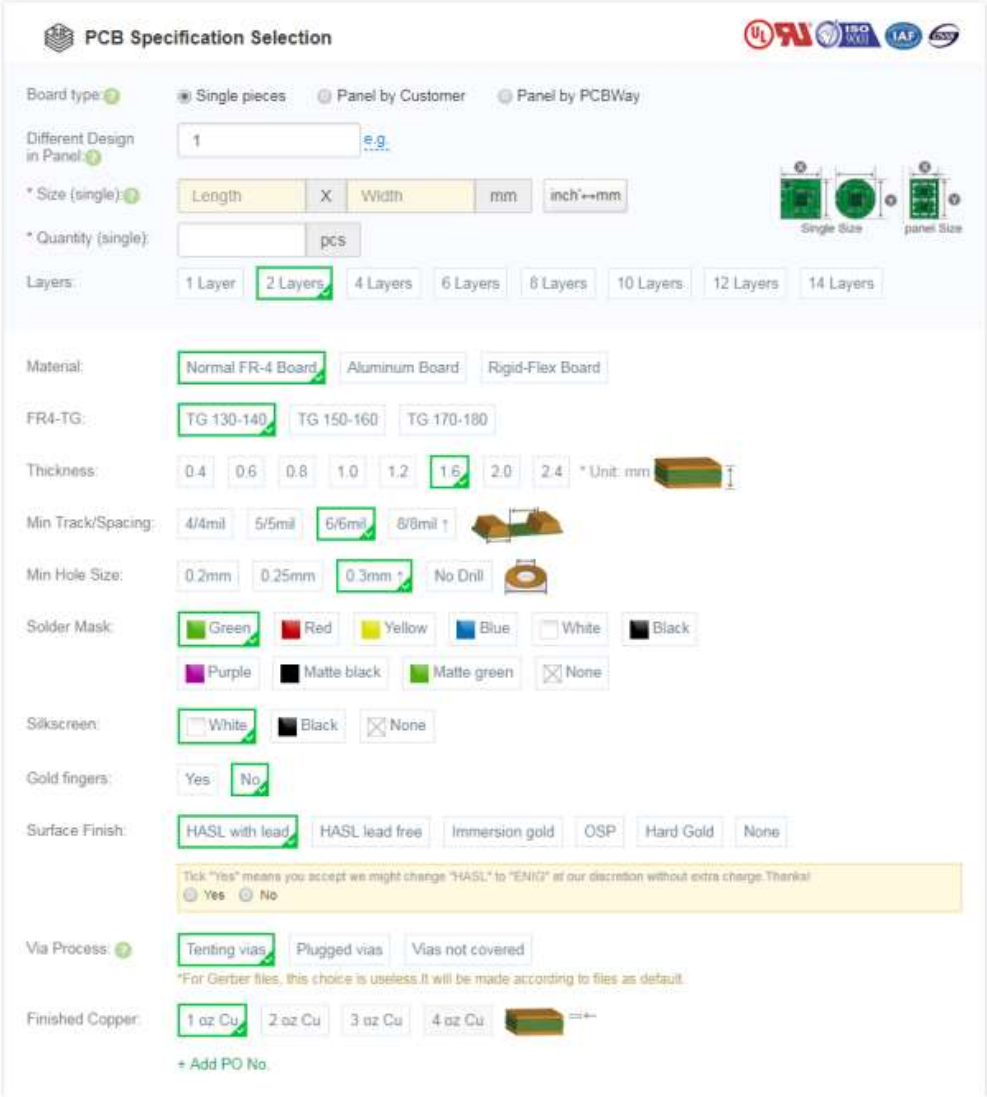
- PCB manufacturers are not checking that your design is functional. You will not get an error to say your diode is the wrong way around, or that you have connected the ground pin to the wrong place on an IC, or that the part you want to use does not support the voltage you are presenting to it etc. They are simply checking your design can be manufactured. It is up to the student(s) to check and double check the schematic is correct.
- PCB manufacturers are not checking track widths. It is up to the student to check and double check the track widths are correct, usually by setting rules in the tool of choice to ensure track widths of that net are above a certain width to carry the proposed current for example.
- PCB manufacturers are not checking footprints. It is up to the student to check and double check datasheets to ensure the footprints are correct.
- Having the board successfully approved by PCBWay, does not automatically mean it will be approved by other manufacturers. All have different capabilities and rules.

PCB manufacturers will generally check that

- All design files are correct and present. i.e. You should get an email to say if the drill file is wrong or not present
- Dimensions of track widths and holes are above the minimum capabilities of the manufacturer.

PCB Submission Form

- Stick with the defaults unless your design warrants extra features.
- Add the length and width of the board
- Minimum quantity is 5 boards
- Cheapest and quickest colour is a green soldermask.
- Tick the yes button to allow HASL to be changed to ENIG. These are just different processes to add a finish to the board.



Typical error showing a missing drill file and no board outline present